

When Does Fusing Hand-Crafted Knowledge With Learned Representations Pay?

(a) Controlled Instrument

Frozen Recipe: SGD 0.1, Cosine, 200 Epochs, Batch 128, Crop and Flip

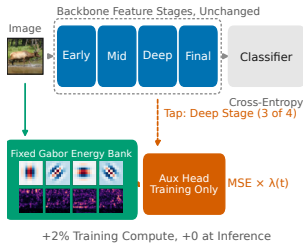
Committed Subset Indices, Per Dataset and Fraction

Pinned Filter Bank (Fingerprinted)

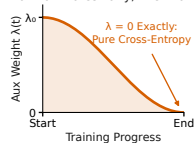
Every Intervention Sees
Byte-Identical Images

Deviations Quarantined, Never Headline

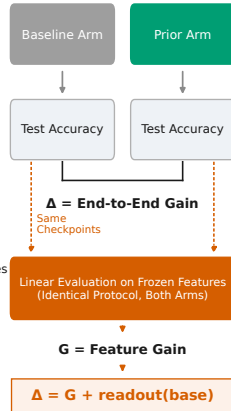
(b) MomentAux: The Fused Prior



Prior Dominates Early, Then Vanishes



(c) Two Measurements



(d) Comparators, By Cost

- ImageNet Transfer External Data
- SimCLR, SimSiam, DINO 2x Compute
- FitNets Learned Teacher 2x Compute
- DeiT Augmentation Recipe Change
- HOG Target 1.02x
- MomentAux (Ours) 1.02x

All Fused Into the Same Frozen Recipe; Combinations Measured Too

(e) When Fusion Pays

